

Electrochemical performance of low temperature fluorinated graphites used as cathode in primary lithium batteries

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Abstract

The present study highlights the electrochemical performance of two series of fluorinated graphites used as the cathode in primary lithium batteries. These compounds were prepared under fluorine gas at room temperature using a catalytic atmosphere made of boron or chlorine fluoride, and then thermally treated between 100 and 600 °C. The electrochemical properties are correlated to a complete physico-chemical characterization, already performed by XRD, NMR, FT-IR and EPR. In particular, important parameters are taken into account: C–F bonding, carbon hybridization, fluorine content (i.e. F/C ratio) and amount of intercalated catalyst residues. It is shown that the average discharge potential of fluorinated graphite used in primary lithium batteries can be predicted owing to the chemical shift values ($\delta_{\text{C-F}}$) obtained by solid ^{13}C NMR. On the other hand, the higher capacity values are achieved for low temperature fluorinated graphite treated at the highest temperatures, i.e. for high fluorination level. The electrochemical performance study of these materials is completed by the study of the effect of simulated storage. The differences between the various samples during electrochemical tests and those observed using different electrolytes are discussed. Fluorinated graphites obtained with a chlorine catalyst or post-treated at temperatures higher than 450 °C are unaffected by ageing.

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1. Introduction

Since the beginning of the 1970s, graphite fluorides have been used for energy storage. Graphite forms several compounds with fluorine which are used as the cathode in primary lithium batteries. These materials can be prepared using different methods. At high temperature (600 °C), fluorine gas reacts with graphite to yield graphite fluoride of general formula $(\text{C}_x\text{F})_n$ [1,2]. This compound exhibits purely covalent C–F bonding and a puckering of the graphene layers, i.e. carbon atoms are sp^3 hybridized [2–4]. A high fluorination level is then obtained, with x close to one; this results in high capacities during electrochemical discharge when these materials are used as primary lithium

battery electrodes. However the faradic yield, lower than 100%, suggests the presence of electrochemically inactive defects such as CF_2 groups which were identified by Infra-red spectroscopy [3].

Another way to prepare graphite fluoride consists in a room temperature synthesis: a gaseous, catalytic atmosphere of F_2 , HF and a volatile fluoride MF_n such as IF_5 , BF_3 , ClF_y ... is then used [5]. For such compounds, the fluorination level strongly depends on the nature of the catalyst fluorides [6]. Nominal compositions of $\text{CF}_{1.03}$, $\text{CF}_{0.47}$ and $\text{CF}_{0.42}$ are obtained for IF_5 , BF_3 and ClF_y catalysts, respectively [7–10]. As the C–F bonding exhibits a semi-covalent character, the bonding energy is weaker than in conventional covalent graphite fluoride $(\text{CF})_n$. Moreover, semi-covalent C–F bonding allows the sp^2 carbon hybridization of the original graphite to be preserved [7–10]. However, the presence of residual catalysts (IF_5 , IF_6^- , BF_4^- , ClHF^- , FHF^-), which are still intercalated

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Table 1
Chemical composition and name of some fluorinated graphites

Inorganic fluoride	Composition of CF _x obtained at RT	Treatment temperature (°C)	Chemical composition after thermal treatment	Name
BF ₃	CF _{0.47} ± 0.01 (called A)	400	CF _{0.85} ± 0.02	A(400)
		550	CF _{1.01} ± 0.02	A(550)
ClF _y	CF _{0.42} ± 0.01 (called B)	400	CF _{0.51} ± 0.01	B(400)
		550	CF _{0.94} ± 0.02	B(550)

in low amounts into the graphitic intervals, significantly changes the electrochemical properties of the samples. This effect is increased by prolonged storage. Indeed, a recent study of Giraudet et al. has shown that fading can occur for graphite fluorides prepared at room temperature using IF₅ as catalyst and can be avoided by both a fluorination post-treatment at a temperature higher than 400 °C, and the choice of an adapted electrolyte [10].

A thermal treatment under pure fluorine gas, performed in the 150–600 °C temperature range for these room temperature graphite fluorides, results in a progressive conversion from semi-covalent into covalent (C_xF)_n. The evolution of the physico-chemical properties as a function of treatment temperature has been carefully investigated by several adapted techniques, such as nuclear magnetic resonance (¹⁹F, ¹³C, ¹¹B and ¹H), electron paramagnetic resonance, X-ray diffraction, FT-infrared spectroscopy. Below 400 °C, the planarity of the sheets is maintained and residual catalysts are progressively removed from the fluorographite matrix [8–10]. Between 400 and 600 °C, layer puckering occurs resulting in the improvement of the fluorination content, which reaches a value close to one. ¹⁹F and ¹³C NMR studies have revealed a progressive evolution of the C–F bonding from semi-covalent to covalent [7–11].

In this study, electrochemical properties of two fluorinated graphites prepared with different catalysts (BF₃ and ClF_y) were investigated and compared to both fluorinated graphite prepared with iodine fluorides (CF_x–IF₅), and high temperature graphite fluoride (CF₁(HT)). These electrochemical performances are discussed as a function of the treatment temperature by considering the physico-chemical properties in order to explain their large evolution. Moreover, the operating limits of these materials as primary lithium battery electrodes will be determined by a study of lithium cell ageing.

2. Experimental

2.1. Sample preparation

Fluorinated graphites were prepared using Madagascar natural graphite powder (particle size of 7.5 µm) at room temperature using synthesis methods already described elsewhere [7–9] which utilize a gaseous mixture of fluorine (F₂), hydrogen fluoride (HF) and an inorganic fluoride (MF_n). Two different MF_n catalysts were used: BF₃ [8]

and ClF_y [9] (synthesis through heating of KCl). The fluorinated graphites thus obtained were called A and B, for BF₃ and ClF_y catalysts, respectively.

Table 1 summarises the chemical composition of the various materials. When the IF₅ catalyst is used, a high fluorination level (CF_{1.03}) is obtained, whereas with BF₃ and ClF_y, the fluorine content is lower (CF_{0.47} and CF_{0.42}, respectively). This difference is related to the Lewis acidity of the fluoride catalyst [6]. The fluorination levels have been calculated by considering the weight uptake. In order to improve the fluorination level and to remove the residual catalyst, these two compounds were heat treated during 3 h under a stream of pure fluorine gas at a temperature (*T*_T) ranging between 150 and 600 °C. Two series of compounds were then prepared, denoted A(*T*_T) and B(*T*_T).

2.2. Characterization

¹³C Magic Angle Spinning (MAS) NMR experiments (10 kHz spinning rate) were carried out at room temperature on a Bruker AVANCE DSX 300 spectrometer operating at 75.47 MHz. The external reference was tetramethylsilane (TMS).

2.3. Electrochemical study

Composite electrodes were composed of fluorinated graphite (about 80% by weight, w/w), graphite powder (10%, w/w) to ensure electronic conductivity and polyvinylidene difluoride (PVDF, 10%, w/w) as a binder. After stirring in propylene carbonate (PC), the mixture was spread uniformly onto a stainless steel current collector disk of 10 mm in diameter. For PC evaporation, the disk was heated at 150 °C in an atmosphere of air and then in a vacuum at 150 °C for 1 h in order to remove traces of water and solvent. The mass of active material was included between 2 and 10 mg. A two electrodes cell was used (Swagelok cell type), where lithium was both the reference and the counter electrode. A PVDF microporous film wet with electrolyte, which was composed of a lithium salt (LiClO₄ or LiBF₄) dissolved in PC (doubly distilled) with 1 mol L^{−1} concentration, was sandwiched between the composite electrode and a lithium metal foil. The cells were assembled in an argon filled glove box. Relaxation was performed for at least 5 h until OCV stabilization. Galvanostatic discharges were performed at room temperature between the initial OCV and 1.5 V with a VMP2–Z under current den-

sity of 10 mA g^{-1} . The capacity Q was measured with a potential cut-off of 2.0 V and the average discharge potential $E_{1/2}$ corresponds to the potential obtained at half the discharge. For each material, 3 tests were systematically performed to check the reproducibility. Ageing was simulated by maintaining the cells at 60°C for 5 days before the electrochemical measurements. Four fluorinated graphites are studied after ageing: A(400), A(550), B(400) and B(550).

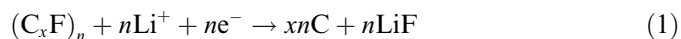
3. Results and discussion

3.1. Electrochemical characteristics

Regardless of the sample and the series A(T_T) or B(T_T), the voltage abruptly drops at an early stage of discharge; it then increases steadily to the main voltage plateau after a few minutes. This behaviour is exemplified in Fig. 1, which shows the galvanostatic discharge curves of the B(T_T) series. The initial voltage drop is characteristic of $\text{Li}/(\text{C}_x\text{F})_n$ cells and is attributed to the low electrical conductivity of the starting material/ $(\text{C}_x\text{F})_n$ [12,13]. The OCV results from a mixed potential between the potential of the intercalated residual species and that of $(\text{C}_x\text{F})_n$, the latter potential being related to the covalent character of the C–F bond [14]. Indeed the OCV of inorganic fluorides–GIC (MF_n –GIC) is greater than 4 V [15]. Hence, the OCV can be correlated with both the amount of residual catalyst and the covalent character σ of the C–F bond. The compounds A and B possess the same bonding as indicated by the similar ^{13}C chemical shift of the C–F groups, 78.5 and 78 ppm for A and B, respectively [8,9]. So, the difference between the OCV values (3.67 and 3.18 V, for A and B, respectively) is explained by the highest amount of residual species for A as indicated by NMR. On the other hand, the OCV (3.18 V) of CF_x – IF_5 is probably due to a more covalent C–F bond ($\delta \approx 84 \text{ ppm}$). Regardless of which graphite fluoride is studied, a progressive decrease of the catalyst residue content occurs when the samples are treated at a temperature lower than 400°C . Simultaneously to this de-intercalation, the covalency of the C–F bond increases

with the treatment temperature [7–11]. This results in a decrease of the OCV value.

For fluorinated graphite, the theoretical capacity is linked to the electrochemical reduction reaction during galvanostatic discharge (Eq. 1).



This reaction consists in the C–F bond dissociation through LiF formation. The higher the fluorine content, the higher the capacity. A(T_T) and B(T_T) series are similar in terms of evolution in fluorine content. First, for the A(T_T) series, increasing the treatment temperature results in the improvement of the fluorination content (Fig. 2). As expected, the increase in capacity is then related to the fluorine content and the capacity varies between 310 Ah kg^{-1} , for fluorinated graphite A ($\text{CF}_{0.47}$) and 890 Ah kg^{-1} for A(550) ($\text{CF}_{1.01}$). The latter value is higher than the capacity of conventional covalent graphite fluoride (820 Ah kg^{-1}) [16]. For the samples treated at temperatures included between 150 and 250°C , the capacities are lower than for the raw compound A. Post-treatment at temperatures lower than 300°C may convert fluorine trapped during the synthesis of the low temperature graphite fluoride, into mobile F^- . Indeed, after such treatment, a higher reactivity with air moisture (weight uptake) has been registered for samples post-treated at temperatures ranging between 150°C and 250°C in comparison to the untreated sample A [17]. Mobile F^- reacts with air to form HF molecules as revealed by ^{19}F NMR [8]. So, the decrease in capacity can be mainly explained by the mass of the F^- species taken into account in these calculations, whereas these ions do not actually participate in the lithium battery electrochemistry.

The highest voltage plateau of the A(T_T) series is observed for the compound A (3.13 V) and can be correlated with the highest C–F bond ionicity (Fig. 3). In the 250 – 550°C temperature range, the covalent character of the C–F bonding increases with treatment temperature. This is indicated by the evolution of the ^{13}C NMR chemical shift of the carbon atoms which are in strong interaction with fluorine ($\delta_{\text{C-F}}$): the higher the chemical shift,

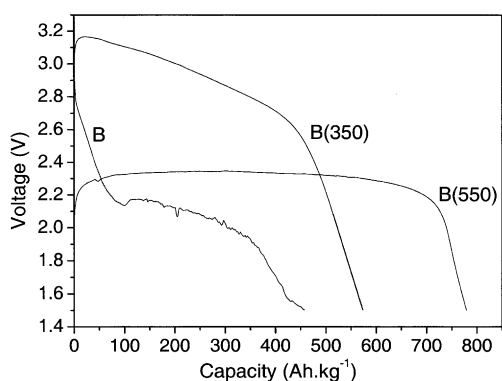


Fig. 1. Galvanostatic discharge curves of some fluorinated graphites B(T_T).

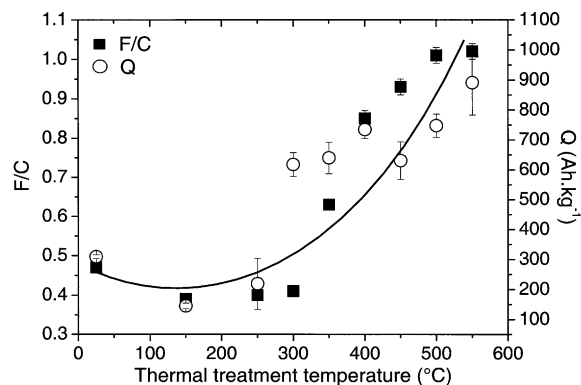


Fig. 2. F/C ratio (■) and capacity (○) of fluorinated graphites A(T_T) as a function of thermal treatment temperature.

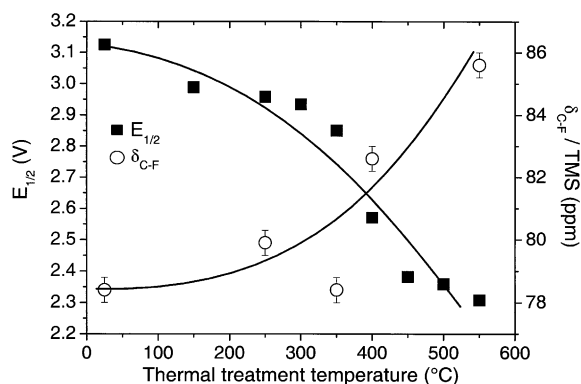


Fig. 3. Average discharge voltage $E_{1/2}$ (■) and ^{13}C chemical shift of the C–F resonance (○) of fluorinated graphites A(T_T) as a function of thermal treatment temperature.

the higher the C–F bond covalency. The increase in C–F bond covalency results in a decrease in discharge potential because covalent C–F bonds require more energy to be broken during the discharge. In comparison, purely covalent $\text{CF}_1(\text{HT})$ exhibits a lower discharge voltage, equal to 1.96 V [16] or to 2.1 V in our electrochemical conditions.

The galvanostatic discharge curve of B (Fig. 1) is typical of a bad working cathode as observed in F–GIC containing a high amount of intercalated catalyst [18]. For treatment temperatures below 300 °C, the same phenomenon disturbs the electrochemical discharge. Between 300 and 400 °C, the capacities are low (Fig. 4). This is mainly related to the low fluorination level. However, the presence of the catalyst species can decrease the electrochemical performances because: (i) their mass is considered to constitute part of the active mass whereas they do not participate (or slightly) in the electrochemistry of the lithium battery, (ii) together with HF molecules, they can form a complex [6] which may hinder lithium's accessibility to active C–F sites within the layer. As the F/C ratio increases with the treatment temperature, the capacity increases as a function of treatment temperature. The highest capacity is equal to 739 Ah kg^{-1} for B(550). Nevertheless, the F/C ratio of B(550) does not reach the theoretical value of 1

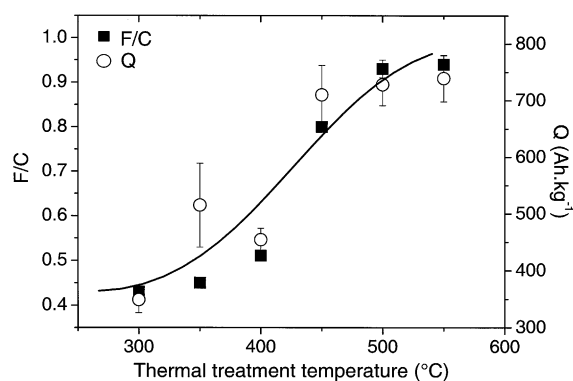


Fig. 4. F/C ratio (■) and capacity (○) of fluorinated graphites B(T_T) as a function of thermal treatment temperature.

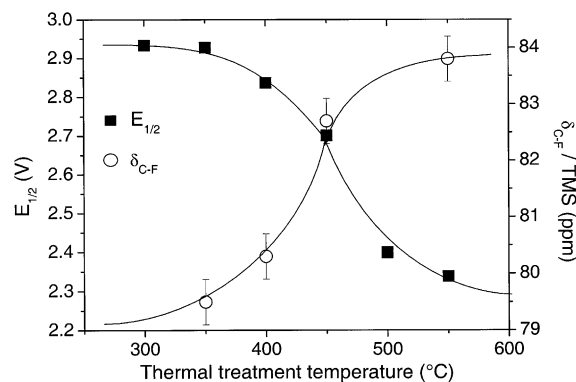


Fig. 5. Average discharge voltage $E_{1/2}$ (■) and ^{13}C chemical shift of the C–F resonance (○) of fluorinated graphites B(T_T) as a function of thermal treatment temperature.

(F/C = 0.94) because of the presence of structural defects induced during fluorination. Moreover the presence of electrochemically inactive CF_2 and CF_3 groups (related to structural disorder and detected by ^{19}F and ^{13}C NMR [9]) leads to a lower capacity in comparison to the theoretical one of B(550) (844 Ah kg^{-1}).

As for the A(T_T) series, the increase in chemical shift $\delta_{\text{C-F}}$ of the B(T_T) series with the treatment temperature indicates a more covalent C–F bonding leading to a decrease in discharge voltages (Fig. 5). More generally, materials with close chemical shifts $\delta_{\text{C-F}}$ have the same average discharge potential regardless of the catalyst used during the synthesis of the raw low-temperature-fluorinated graphite.

Irrespective of the choice of catalyst, capacities are not always proportional to the F/C content as expected. Fig. 6 clearly shows that for $\text{CF}_x\text{--IF}_5$, although the fluorination level is nearly constant, the capacities of post-treated $\text{CF}_x\text{--IF}_5$ increase with the treatment temperature. These improvements are then rather correlated with the removal of the iodine fluoride species which occurs progressively during the thermal treatment. Capacities are in fact inversely proportional to the catalyst content indicating the strong correlation between these two parameters. To summarize, removal of the intercalated species is neces-

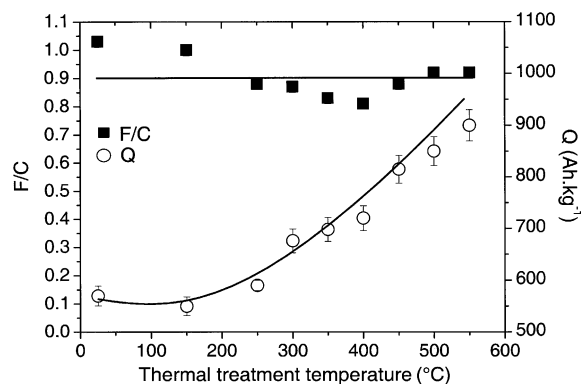


Fig. 6. F/C ratio (■) and capacity (○) of fluorinated graphites $\text{CF}_x\text{--IF}_5$ as a function of thermal treatment temperature.

sary to enhance electrochemical performance. So, in terms of future commercial uses, the most interesting samples are those that have undergone refluorination at temperatures higher than 400 °C. For these electrode materials, it is then important to investigate the behaviour of the lithium cells upon prolonged storage.

3.2. Galvanostatic discharges upon ageing

Ageing tests are simulated by maintaining the lithium/fluorinated graphite cell at 60 °C for 5 days. The electrochemical characteristics (average discharge voltage, capacity and specific energy) of some fluorinated graphite/lithium batteries, studied before and after ageing, are summarized in Table 2.

The ageing of the fluorinated graphite A(400), prepared with BF_3 and treated at 400 °C, leads to a drastic capacity decrease when LiClO_4 is used as the electrolyte salt. In this case, the fluorographite matrix can be stabilized during the electrochemical tests by using an electrolyte containing BF_4^- since no significant modification of the capacity appears. Such an effect can be initially correlated with the presence of the same anion, BF_4^- , initially trapped in the fluorocarbon matrix and in the electrolyte salt. However, in the case of $\text{CF}_x\text{-IF}_5$, ageing is also lessened when LiBF_4 is used instead of LiClO_4 [10]. Since solvent co-intercalation occurs in the potential range under study [19], salt-solvent interactions must also be considered to fully explain these evolutions. Complementary studies, e.g. electrochemical impedance spectroscopy, must be carried out to investigate this hypothesis.

Another explanation for the capacity decrease has been previously shown in the case of fluorinated graphites prepared with IF_5 : the electrolyte can be decomposed; this reaction occurring upon ageing involves the residual catalysts. The presence of decomposition products on the electrode surface and/or within the host matrix could hinder the lithium diffusion into the fluorographite matrix [19]. In the case of A(400), when $\text{LiBF}_4\text{-PC}$ and $\text{LiClO}_4\text{-PC}$

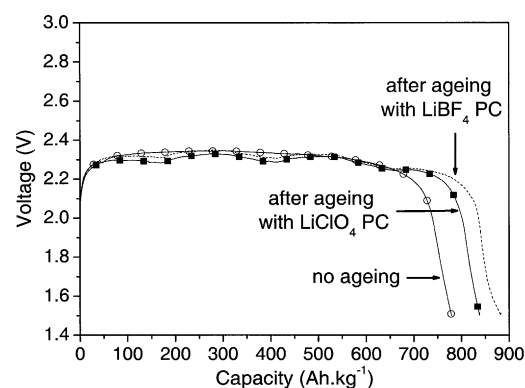


Fig. 7. Galvanostatic discharge curves of fluorinated graphite B(550) before and after ageing test.

electrolytes are used, similar processes can occur upon ageing explaining the decrease in capacities.

For A(550), with no residual catalyst, no decrease in capacity is observed when the two electrolytes constituted of LiBF_4 or LiClO_4 salts are used, confirming our initial hypothesis about the strong effect of the intercalated species. As a matter of fact, in this sample, no residual catalyst is present which can react with the cell. Moreover, the fluorographite matrix of A(550) seems to be stable. Specific energy (Table 2) is unaffected by ageing revealing good electrochemical performances for A(550). B(550) exhibits similar behaviour (Fig. 7). In comparison, fluorinated graphites prepared with IF_5 and treated under fluorine at 530 °C are also unaffected by ageing [10].

Contrary to A(400), no effect of ageing is observed for B(400) even if the compound contains intercalated residual catalysts, showing that the nature of the intercalated species also plays a key role. The average discharge voltage is similar before and after ageing suggesting no degradation of the fluorinated graphite and of the cell components. So, specific energy is maintained after simulated ageing. Although the fluorinated graphites B(T_T) have a lower specific energy in comparison to A(T_T), their electrochemical performance after ageing is better.

4. Conclusion

As the nature and the content of intercalated species change (BF_4^- and FHF^- for A(T_T), ClHF^- and FHF^- for B(T_T), IF_6^- and IF_5 for $\text{CF}_x\text{-IF}_5$), their reactivity vis-à-vis the electrolyte are different. It is difficult to generalize but this study allows experimental classification of graphite fluorides as electrode material for primary lithium batteries.

The higher the C–F covalency, the lower the average discharge potential. Moreover, the average discharge potential of fluorinated graphite used in primary lithium batteries can be predicted owing to chemical shift values ($\delta_{\text{C-F}}$) obtained by solid ^{13}C NMR. Indeed, materials with close chemical shifts have the same average discharge potential independent of the catalyst used during the syn-

Table 2

Discharge potential, capacity and energy of some fluorinated graphites for different experimental conditions

	A(400)	A(550)	B(400)	B(550)
No ageing				
Q (Ah kg $^{-1}$)	787 ± 31	941 ± 68	501 ± 34	779 ± 43
$E_{1/2}$ (V)	2.60	2.33	2.80	2.34
E (Wh kg $^{-1}$)	2046 ± 81	2193 ± 158	1403 ± 95	1819 ± 100
After ageing with LiClO_4PC				
Q (Ah kg $^{-1}$)	430 ± 21	955 ± 103	545 ± 27	838 ± 54
$E_{1/2}$ (V)	2.41	2.36	2.79	2.29
E (Wh kg $^{-1}$)	1037 ± 51	2254 ± 243	1509 ± 75	1918 ± 124
After ageing with LiBF_4PC				
Q (Ah kg $^{-1}$)	807 ± 27	852 ± 74	581 ± 33	882 ± 57
$E_{1/2}$ (V)	2.44	2.16	2.78	2.32
E (Wh kg $^{-1}$)	1971 ± 66	1843 ± 160	1615 ± 92	2046 ± 132

thesis of the raw low-temperature-fluorinated-graphite. The higher the fluorination level, the higher the capacity. However, capacity values can be slightly decreased due to the amount of residual catalyst.

The most important effect of the intercalated species occurs for the ageing study. Indeed, these species are deintercalated and/or reduced during the discharge and move into the cell. They can react with the cell components, such as the current collector and the electrolyte, which leads to cell degradation. This process particularly affects the sample treated at low temperature, indicating a strong effect of the intercalated species BF_4^- , IF_n (IF_5 and IF_6^-), HF_2^- . Only fluorinated graphites obtained with a chlorine catalyst or post-treated at temperatures higher than 450 °C for which no intercalated species are still trapped in the fluorographite matrix, are unaffected by ageing. To take the study on ageing mechanisms further, changes at the electrode/electrolyte interface and kinetic aspects, e.g. lithium ion diffusion and electrolytic conductivity must be studied by EIS. Moreover, the use of mixed solvents such as PC–ethylene carbonate (EC) resulted in a drastic decrease in the effects of ageing in the case of fluorinated graphite obtained at room temperature with IF_5 [10]. Similar electrolyte should be tested for ageing study of fluorinated graphite obtained at room temperature with BF_3 .

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